

EduGenie: Google Gemini Powered Learning Assistant

Project Documentation & Implementation Report

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1. Project Overview

EduGenie is an AI-powered educational assistant designed to make self-study interactive, structured, and personalized. Built on FastAPI and powered by Google Gemini, EduGenie supports learners with direct question answering, intuitive topic breakdowns, automated quiz generation, text summaries, and phased study paths.

2. Architecture & File Structure

The project follows a clean asynchronous structure separating environment configurations, Pydantic schemas, modular AI endpoints, and client-side web templates:

Visual Studio Code – EduGenie Workspace ExplorerPython 3.11

EDUGENIE-AI

> .venv/

> __pycache__/

> static/ (style.css)

> templates/ (index.html)

≡ .env

≡ .gitignore

⚙ config.py

⚙ schemas.py

🔗 main.py

🔗 explanation_module.py

🔗 qna.py

🔗 quiz_module.py

🔗 summary_module.py

🔗 learning_path.py

≡ requirements.txt

main.py - Asynchronous FastAPI REST Engine

from fastapi import FastAPI, Request

from config import settings

from schemas import ExplainRequest, QuizRequest, SummaryRequest

app = FastAPI(title="EduGenie", description="Google Gemini Powered Learning Assistant")

Endpoint: Concept explanation with pedagogical heuristics

@app.post("/explain")

async def explain(payload: ExplainRequest):

return {"result": explain_topic(payload.topic, payload.level)}

3. Technology Stack

Backend Framework: FastAPI (v0.110+)	Server: Uvicorn (ASGI v0.28+)
Data Validation: Pydantic & Pydantic-Settings	AI Model: Google Gemini API (gemini-1.5-pro / flash)
Templating: Jinja2 (HTML5/CSS3)	Local Fallback: HuggingFace / PyTorch (LaMini-Flan-T5)

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4. Milestones & Implementation Details

Milestone 1: Environment & Settings

Environment configurations and API credentials are systematically managed through `config.py` using Pydantic Settings, securely loading sensitive variables such as the Gemini API key from a local `.env` file.

```
.env – Environment Configuration (VS Code) UTF-8
APP_NAME=EduGenie
GEMINI_API_KEY=AQ.Ab8RN6LvPgtAdrKgI5Epq4-MuRh8ovZLvWgP920q5BJLbN4jKg
GEMINI_MODEL=gemini-3.8-flash
LOCAL_EXPLAINER_ENABLED=false
LOCAL_EXPLAINER_MODEL=MBZUAI/LaMini-Flan-T5-783M
MAX_INPUT_CHARS=20000
REQUEST_TIMEOUT_SECONDS=90
```

Milestone 2: Schemas & Core Modules

The data interface layer in `schemas.py` defines robust Pydantic models for request payloads (`ExplainRequest`, `QAResponse`, `QuizRequest`, `SummaryRequest`, `LearningPathRequest`) and their corresponding output models to guarantee strict type safety.

- **`explanation_module.py`**: Breaks down concepts by difficulty level (beginner, intermediate, advanced).
- **`qna.py`**: Structured question answering logic powered by Google Gemini SDK.
- **`quiz_module.py`**: Dynamic MCQ generator with regex JSON sanitization to clean markdown backticks.
- **`summary_module.py`**: Educational summarization condensing text into concise takeaways.
- **`learning_path.py`**: Phased multi-week learning roadmaps from Foundations to Independent Projects.

Milestone 3: FastAPI Routing

The application exposes structured endpoints in `main.py` to handle client operations:

- **GET** `/`: Web dashboard UI
- **GET** `/health`: Status health check
- **POST** `/qa`: Question answering
- **POST** `/explain`: Topic explanation
- **POST** `/quiz`: Quiz generation
- **POST** `/summarize`: Text summarization
- **POST** `/learn/recommendations`: Personalized study path

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Milestone 4: User Interface

The dynamic frontend is powered by Jinja2 rendering templates/index.html. It provides an intuitive web interface containing interactive task selectors, flexible user prompt inputs, and real-time response viewports.

Google Chrome – index.html (EduGenie Personal Learning Studio) 127.0.0.1:8000

← → ↻ http://127.0.0.1:8000

PERSONAL LEARNING STUDIO

EduGenie

Turn difficult material into clear explanations, useful summaries, quizzes, and a plan you can actually follow.

Ready to learn

Mode: Summarize text ▾ Level: beginner ▾

Your material: explain photosynthesis

Quiz questions: 5 Weeks: 4

Generate Done.

Result: Photosynthesis is the process plants use to make food. It uses sunlight, water, and carbon dioxide. Clear

5. Run & Verification

- **Install dependencies:** `pip install -r requirements.txt`
- **Run server:** `uvicorn main:app --reload`
- **Verify:** Access via web browser and confirm system operational state via /health endpoint.

```
PS D:\Documents\EduGenie-Ai> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['D:\\Documents\\EduGenie-Ai']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [17749] using StatReload
INFO: Started server process [17751]
INFO: Application startup complete.
INFO: 127.0.0.1:8000 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:8000 - "POST /explain HTTP/1.1" 200 OK
```

6. Future Scope

Future expansions for EduGenie will introduce voice query processing, native multi-modal document parsing (PDFs, diagrams, and hand-written notes), automated learning progress analytics, and integration with major Learning Management Systems (LMS) such as Canvas and Moodle.

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Internal College Record